PORTFOLIO PROJECT · AI & AUTOMATION

AI-Driven Project Intake & Automated Resource Dispatcher

A multi-phase intelligent automation system that handles an entire construction project from the moment a form is submitted — organising documents, extracting data using AI, matching the right engineer to the right project, assigning tasks, calculating deadlines, and notifying the team. All automatically.

What it does	Automates project intake, document organisation, estimator assignment, deadline setting, and team notification
Tools used	n8n · Google Drive · Google Sheets · Google Gemini AI · Monday.com · Slack · JavaScript
Who benefits	Construction managers, estimation teams, and operations looking to eliminate manual admin
Key features	Automatic folder creation · Recursive estimator availability loop · Cross-platform sync · Slack notifications
Built by	Sydney Rey De Leon · AI & Automation Specialist | Registered Civil Engineer

What Is This Project?

In construction, every new project starts the same way: someone submits a set of documents — plans, technical requirements, specifications — and then the manual work begins. Someone has to organise the files, read through the documents, figure out what the project needs, find a qualified engineer who's actually available, assign the work, set a deadline, and tell the team. That process is slow, repetitive, and full of opportunities for things to fall through the cracks.

This system eliminates that entire manual chain. From the moment a project form is submitted, an intelligent multi-phase automation takes over. It organises the documents, uses **Google Gemini AI** to read and extract the data, continuously monitors staff availability until the right person is free, assigns the project in **Monday.com**, calculates the due date, updates **Google Sheets**, and sends the full project briefing to the team via **Slack**. **No emails back and forth. No manual assignment. No missed deadlines.**

3	0	Auto	Live
Pipeline Phases	**Manual Steps**	**Assignment Loop**	**Status Tracking**
Intake, Resource Allocation & Handoff	Fully automated from form to team notification	Keeps searching until a free estimator is confirmed	Monday.com & Sheets updated in real time

> Why a Civil Engineer built this: Having worked across project engineering, QS, QC, and TBM operations, the pain of manual project intake and resource coordination is deeply familiar. This system was designed with that real-world construction workflow in mind — not as a generic automation, but as a purpose-built solution for how construction estimation teams actually operate.

The 3 Phases — System Overview

The system is built in three connected phases. Each phase hands off cleanly to the next, forming one continuous automated pipeline:

Phase 1 · Project Intake & Infrastructure

Captures the form submission, builds the folder structure in Google Drive, sorts documents by type, and uses Google Gemini AI to extract and structure all data from the uploaded files.

Phase 2 · State-Machine & Resource Allocation

Takes the extracted project data and enters a smart matching loop — continuously checking staff availability until the right engineer is found and confirmed. No project sits waiting with no owner.

Phase 3 · Synchronisation & Team Handoff

Assigns the project in Monday.com, calculates the due date using JavaScript, syncs everything to Google Sheets, and sends a complete project briefing to the right person on Slack.

PHASE 1 · PROJECT INTAKE & INFRASTRUCTURE

From Form Submission to Structured AI-Ready Data

Everything begins with a single form submission. From that moment, the system takes full control of organising, categorising, and preparing all project documents — so the estimation team receives clean, structured data instead of a pile of raw files.

1

A project form is submitted

The process is triggered the moment someone fills out and submits the project intake form. This submission carries all the initial project details — project name, client information, and any uploaded documents such as floor plans, technical drawings, or specifications.

2

A dedicated Google Drive folder is created automatically

Without any manual action, the system immediately creates a structured folder in Google Drive specifically for this project. Sub-folders are generated inside it to separate different document types — for example, one folder for technical requirements and another for plans. Everything has a home from the very start.

3

Documents are sorted and uploaded into the right folders

The uploaded files are read and categorised by the system. Technical requirement documents go into one sub-folder. Plan files go into another. Each file is uploaded to its correct location automatically — no manual sorting needed.

4

The Master List in Google Sheets is updated

Once the files are organised, the system tallies the document count and records the project details in the Master List spreadsheet. This acts as the central registry for all projects — always up to date, always accurate, with no manual data entry required.

5

Google Gemini AI reads and extracts data from the uploaded files

This is where the intelligence kicks in. Google Gemini AI — Google's advanced AI model — is given access to the uploaded documents and instructed to read through them. It identifies and extracts the key data points: project scope, material requirements, quantities, specifications, and anything else relevant to the estimation process.

6

Unstructured documents become structured, usable data

Raw files — PDFs, drawings, specification documents — are transformed by the AI into structured JSON format. Think of it as the AI reading a complex blueprint and converting it into a clean, organised list of information that the rest of the system can actually work with. What used to take an estimator hours to read and summarise is done in seconds.

Why this matters: Estimation teams spend a significant portion of their time just reading through documents and manually pulling out numbers. By the time Phase 1 is complete, all that work has already been done by the AI — and the estimator can go straight to analysis and decision-making.

PHASE 2 · ESTIMATOR AVAILABILITY & RESOURCE ALLOCATION

Automatic Loop Until the Right Person is Found

Once the project data is structured and ready, the system needs to find the right person to work on it. But people aren't always available — they're on other projects, on leave, or at capacity. This phase handles that challenge with a smart, persistent matching loop that keeps checking until a match is confirmed.

7 Project data is cleaned and prepared for matching

The structured data from Phase 1 is normalised — meaning any inconsistencies, formatting differences, or gaps are cleaned up. If multiple projects came in at once, they are split into individual batches so each one can be processed separately and accurately.

8 The system checks Monday.com for all current project assignments

The automation connects to Monday.com and reads the current state of the project board — which engineers are already assigned to active projects, what their workload looks like, and which projects are still pending without an owner.

9 Staff availability is checked against the employee database

The system pulls the full employee list and checks each person's current status. It identifies who is available, who is currently occupied, and who has the right skills or role for this type of project. Availability is checked in real time against live data — not a cached snapshot.

10 If no one is available, the system waits and tries again

This is the recursive loop at the heart of Phase 2. If no available engineer is found, the system doesn't fail or send an error — it simply pauses, waits, and then re-checks automatically. This loop continues running in the background until a match is found. A project is never left unassigned simply because everyone was busy at the exact moment it arrived.

11 The right engineer is identified and selected

Once an available, qualified engineer is confirmed, the system locks them in for this project. Their details — name, role, current status — are captured and passed forward to the next phase for assignment.

> Why a recursive loop instead of a simple check: A one-time availability check fails the moment all engineers are busy. The recursive loop ensures that the system keeps trying — so when someone finishes a project and becomes available, the waiting assignment is picked up immediately without anyone having to manually trigger it again.

PHASE 3 · SYNCHRONISATION & TEAM HANDOFF

Assignment, Deadlines, and Instant Team Notification

With the right engineer identified, the final phase connects all the platforms, makes the formal assignment, calculates when the work is due, and puts everything in front of the team — in the tools they already use every day.

12

The project is assigned in Monday.com

The system makes a direct API call to Monday.com and creates or updates the project task — assigning it to the selected engineer, setting the project name, linking the relevant documents, and marking the status as active. The project board updates in real time, giving the whole team instant visibility.

13

The due date is calculated using JavaScript

A JavaScript module runs inside the automation to calculate the project deadline. It takes into account the current date, the project complexity, and any predefined turnaround rules — then outputs a precise due date. This calculated date is then synced back to Monday.com so the timeline is always accurate from the moment of assignment.

14

Google Sheets is updated with the final project status

All assignment details — engineer name, project ID, due date, document links, and current status — are written back to the Master List in Google Sheets. This keeps the spreadsheet in sync with Monday.com and gives anyone on the team a single place to check the status of any project at any time.

15

The Slack user database is queried to find the right recipient

Before sending the notification, the system queries the Slack user directory to find the exact Slack profile of the assigned engineer and any relevant team members. This ensures the message goes to the right person — not just a generic channel — without needing to hard-code anyone's details.

16

A full project briefing is sent via Slack

The assigned engineer and the team receive an automated Slack message containing everything they need to get started: the project name, client details, due date, links to the Google Drive folder with all documents, and a direct link to the Monday.com task. Nothing is left out. Nothing needs to be chased manually.

End result: From a single form submission, the system has created a folder, sorted and read every document with AI, found the right engineer, assigned the project in Monday.com, set the deadline, updated the master spreadsheet, and sent the full briefing to the team on Slack — all without a single manual step.

The Tools Working Together

Seven tools are connected in this system. Each one plays a specific role in the pipeline:

n

n8n — The automation engine

Make is the platform that powers all three phases. It connects every tool, runs the logic, manages the recursive matching loop, and orchestrates the entire workflow from start to finish.

G

Google Drive — The document organiser

Every project gets its own folder structure created automatically. Technical documents and plans are sorted into the right sub-folders the moment the form is submitted.

G

Google Sheets — The master project registry

The central spreadsheet tracks every project — document counts, assignment status, engineer details, and due dates. It's always in sync with Monday.com and updated automatically after every change.

G

Google Gemini AI — The intelligent document reader

Google's AI model reads through uploaded construction documents and extracts structured data. It turns raw plans and specs into usable, organised information that the system can process — in seconds.

M

Monday.com — The project management board

Tasks are created, assigned, and updated in Monday.com via direct API calls. The project board reflects every assignment in real time, giving the whole team live visibility into what's being worked on and by whom.

J

JavaScript — The calculation engine

Custom JavaScript modules run inside Make to handle logic that goes beyond simple automation — specifically, calculating accurate project due dates based on complexity and workload rules.

S

Slack — The team notification channel

Once everything is set up, the assigned engineer receives a complete project briefing directly in Slack — with all document links, the Monday.com task link, and the deadline included. No emails, no manual handoffs.

Who Benefits from This System?

This system was built with three specific groups in mind — each one gets something different out of it:

Construction Managers

Real-time project tracking without checking in with anyone. Every project is visible in Monday.com the moment it's assigned, with a due date already set. No status meetings needed just to find out where things stand.

Estimation Teams

Immediate access to AI-extracted plan data the moment a project arrives. No more spending hours reading through documents manually. The data is already pulled, structured, and ready to work with before the estimator even opens their laptop.

Operations

Drastically reduced administrative overhead. No manual file sorting, no chasing people to confirm availability, no copy-pasting details between platforms. The system handles all of it, eliminating the most time-consuming parts of the intake process entirely.

Why This Matters

✓ **No project ever gets lost in a pile of emails**

Every submission is instantly captured, organised, and tracked in a live system. Nothing waits in an inbox or gets forgotten between handoffs.

✓ **AI does the reading so your team doesn't have to**

Google Gemini extracts data from construction documents in seconds — work that would otherwise take an experienced estimator hours to do manually.

✓ **The right engineer is always found, no matter how busy the team is**

The recursive availability loop means no project ever stalls because everyone was occupied at the wrong moment. The system keeps checking until a match is confirmed.

✓ **Every platform stays in sync automatically**

Monday.com, Google Sheets, Google Drive, and Slack all reflect the same accurate information in real time — with no manual updates needed across any of them.

✓ **The team gets everything they need the moment a project is assigned**

The Slack notification includes document links, the task link, the due date, and the full project brief. The assigned engineer can start immediately without asking anyone for anything.
